

REGRESSION PROJECT

Machine Learning — House Prices Prediction

Regression Analysis • Spring 2026

Alexandria University — Faculty of Computers and Data Science

Team Members

Full Name	Student ID
• Ewan Hany Shaban Mahmoud	2401245838
• Abd Elrahman Eslam Omar	2401242055
• Ali Ibrahim Gaber	2401249410
• Omar Mahmoud Elsayed	2401246509
• Ziad Khaled Abd Elrahman	2401243947
• Mohamed Khaled Fathy	2401249678
• Mohamed Elsayed Ali	2401246442
• Omar Hassan Mohamed	2401241409
• Ammar Yasser Ibrahim	2401246882
• Amr Mostafa Etman	2401246342

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1. Abstract

This report presents a comprehensive implementation and analysis of Linear Regression from scratch using only NumPy, applied to the Ames Housing dataset — a widely-used benchmark containing 1,460 residential property records and 81 descriptive attributes. The central objective of this project is to predict house sale prices accurately while deepening understanding of regression theory through direct implementation rather than reliance on black-box libraries.

Three distinct algorithmic perspectives were developed and compared: (1) Multiple Linear Regression via Gradient Descent (the ML engine), (2) Simple Linear Regression via the Normal Equation (the Numerical engine), and (3) Simple Linear Regression via statistical formulas for slope and intercept (the Statistical engine). Lasso (L1) and Ridge (L2) regularization techniques were integrated into all models to prevent overfitting and support feature selection.

Three model specifications were evaluated: a single-predictor model using GrLivArea (above-grade living area), a full model using all 56 engineered numeric features, and a domain-knowledge model using 10 curated features. Results show that the full model achieves the highest R^2 of 0.7947 on the test set, while the domain knowledge model delivers 0.7750 with far fewer features, demonstrating effective dimensionality reduction. All models were evaluated using MSE, RMSE, and R^2 metrics, and visualized using interactive Plotly dashboards.

2. Introduction

2.1 What Is Regression?

Regression analysis is a fundamental technique in statistics and machine learning used to model the relationship between a dependent variable (the target, or response) and one or more independent variables (the predictors, or features). The goal is to fit a mathematical function to the observed data such that future predictions on unseen inputs are as accurate as possible. In its simplest form, this function is a straight line — hence the name *linear* regression.

The model assumes that the target variable y can be approximated as a linear combination of input features $x_1, x_2, \dots, x_p$ plus a bias term and an error term epsilon:

$$y = B_0 + B_1x_1 + B_2x_2 + \dots + B_px_p + e$$

Here, B_0 is the intercept (bias), B_1 through B_p are the regression coefficients (weights), and e is the residual error. The objective is to estimate these coefficients such that the sum of squared residuals is minimised.

2.2 Types of Regression

- **Simple Linear Regression:** One independent variable, one dependent variable. The model produces a straight line of best fit.
- **Multiple Linear Regression:** Two or more independent variables. The model learns a hyperplane in high-dimensional space.
- **Ridge Regression (L2):** Adds a penalty term proportional to the square of the coefficients, preventing any single weight from growing excessively large.
- **Lasso Regression (L1):** Adds a penalty proportional to the absolute value of the coefficients, encouraging sparsity (driving some coefficients to exactly zero).
- **Polynomial Regression:** Non-linear variant that uses polynomial terms of the original features as additional predictors — still linear in its parameters.

2.3 Importance of House Price Prediction

House price prediction is one of the most economically significant applications of machine learning. Accurate valuation models assist real estate agents in pricing homes competitively, enable banks to assess mortgage risk objectively, and help urban planners understand which neighborhood features drive market value. The Ames, Iowa dataset is a canonical benchmark precisely because it reflects the complexity of real-world pricing — prices are influenced by dozens of interacting features including location, size, quality, age, and amenities.

2.4 Project Objectives

- Implement linear regression from scratch using only NumPy — no sklearn LinearRegression.
- Compare three algorithmic engines: Gradient Descent, Normal Equation, and Statistical formulas.
- Integrate L1 (Lasso) and L2 (Ridge) regularization to handle overfitting.
- Evaluate three model specifications differing in feature set size and selection strategy.
- Build interactive Plotly visualizations to explore data and model behavior.
- Provide a rigorous, academic-quality analysis suitable for university submission.

3. Dataset Description

3.1 Source

The dataset used in this project is the **Ames Housing Dataset**, originally compiled by Dean De Cock (2011) as a modern alternative to the Boston Housing dataset. It describes residential properties sold in Ames, Iowa between 2006 and 2010. The dataset was accessed from the local project directory as `../data/House Prices Dataset.csv`.

3.2 Dimensions and Structure

The raw dataset contains **1,460 rows** (one per property sale) and **81 columns** (80 predictor features + 1 target variable). This was confirmed programmatically:

■ **Code:**

```
df = pd.read_csv('../data/House Prices Dataset.csv')
print('Shape:', df.shape)
```

■ **Output:**

```
Shape: (1460, 81)
```

3.3 Feature Types

Feature Type	Count	Examples
Numeric (integer)	43	GrLivArea, YearBuilt, OverallQual, GarageCars
Numeric (float)	3	LotFrontage, MasVnrArea, GarageYrBlt
Categorical (nominal)	23	Neighborhood, HouseStyle, Foundation, SaleType
Categorical (ordinal)	23	KitchenQual, ExterQual, BsmtQual, PoolQC
Target Variable	1	SalePrice (continuous, USD)

Table 3.1: Summary of feature types in the Ames Housing Dataset

3.4 Target Variable

The target variable is **SalePrice** — the property sale price in US dollars. It is a continuous numeric variable with no missing values. Its distribution is right-skewed, with the majority of homes priced between \$100,000 and \$300,000 and a long tail of luxury properties exceeding \$500,000. This skewness is important context for interpreting model residuals.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is the first critical step in any data science project. It involves summarising the main characteristics of the data, identifying patterns, spotting anomalies, and generating hypotheses to guide the modelling process. In this project, EDA was conducted across several dimensions: structural inspection, missing value analysis, distribution analysis of the target variable, and correlation analysis between features and SalePrice.

4.1 Library Imports

The following libraries were imported to support the entire project pipeline:

■ Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
from sklearn.preprocessing import StandardScaler, OrdinalEncoder
from sklearn.model_selection import train_test_split
from sklearn.impute import SimpleImputer
sns.set_theme()
```

pandas handles tabular data loading and manipulation. **matplotlib** and **seaborn** provide static plotting. **numpy** is the numerical computation backbone. From **sklearn**, only preprocessing utilities are used — the regression models themselves are implemented from scratch. `sns.set_theme()` applies a clean aesthetic to all seaborn plots.

4.2 Data Loading and Initial Inspection

■ Code:

```
df = pd.read_csv('../data/House Prices Dataset.csv')
df.head() # Preview first 5 rows
```

The dataset was loaded directly into a DataFrame. The `head()` call confirms the structure — each row is one property and each column is one attribute. Notable columns visible immediately include: Id, MSSubClass, MSZoning, LotFrontage, LotArea, and the target SalePrice.

■ Code:

```
print('Shape:', df.shape)
```

■ Output:

```
Shape: (1460, 81)
```

This confirms **1,460 houses** and **81 columns** — a manageable but richly featured dataset requiring careful preprocessing.

4.3 Data Types and Info

■ Code:

```
print('Info:')  
df.info()
```

■ Output:

```
<class 'pandas.DataFrame'>  
RangeIndex: 1460 entries, 0 to 1459  
Data columns (total 81 columns):  
# Column Non-Null Count Dtype  
0 Id 1460 non-null int64  
1 MSSubClass 1460 non-null int64  
2 MSZoning 1460 non-null object  
... (38 int/float columns, 43 object columns)
```

The `info()` method reveals that the dataset contains a mix of 38 numeric columns (int64/float64) and 43 string columns (object). Several columns have fewer than 1,460 non-null entries, indicating missing values. `SalePrice` has all 1,460 values present — no rows need to be dropped due to a missing target.

■ Code:

```
print('Describe:')  
df.describe()
```

The `describe()` method provides descriptive statistics (count, mean, std, min, quartiles, max) for all numeric columns. Key observations: `GrLivArea` ranges from 334 to 5,642 sq ft; `OverallQual` is rated 1–10; `YearBuilt` spans 1872 to 2010; `SalePrice` ranges from \$34,900 to \$755,000 with mean ~\$180,921.

4.4 Missing Values Analysis

■ Code:

```
missing = df.isnull().sum()  
missing = missing[missing > 0].sort_values(ascending=False)  
print('Missing Values:')  
print(missing)  
missing.head(10).plot(kind='bar')  
plt.title('Top Missing Values')  
plt.show()
```

■ Output:

```
Missing Values:  
PoolQC 1453  
MiscFeature 1406  
Alley 1369  
Fence 1179  
MasVnrType 872
```

```

FireplaceQu 690
LotFrontage 259
GarageType 81
GarageYrBlt 81
GarageFinish 81
GarageQual 81
GarageCond 81
BsmtExposure 38
(19 columns with missing values total)

```

This step computes the count of null values per column, filters to only those with at least one null, and sorts them in descending order. The bar chart makes the pattern visually clear. Key insights:

- **PoolQC (99.5% missing)**: Nearly all houses lack a pool — these NaNs likely represent 'no pool', not unknown data.
- **MiscFeature (96.3%), Alley (93.8%), Fence (80.8%)**: Similarly, these represent absent features, not truly missing information.
- **LotFrontage (17.7%)**: A meaningful numeric feature — requires imputation strategy (median used later).
- **GarageYrBlt, GarageFinish, etc. (5.5%)**: Garage-related nulls likely correspond to houses without a garage.

4.5 Target Variable Distribution

■ Code:

```

plt.hist(df['SalePrice'], bins=30)
plt.title('SalePrice Distribution')
plt.xlabel('Price')
plt.ylabel('Frequency')
plt.show()

```

The SalePrice histogram reveals a right-skewed (positively skewed) distribution. The bulk of houses are priced between \$100,000 and \$300,000, with a long right tail extending to \$755,000. This skewness is typical of real-world housing markets where a minority of premium properties occupy the high end.

Figure 4.1: Histogram of SalePrice — note the right-skewed distribution indicating a majority of affordable homes with a tail of high-value properties.

4.6 Correlation Analysis

■ Code:

```

corr = df.corr(numeric_only=True)
top_corr = corr['SalePrice'].sort_values(ascending=False)
print('Top 20 Correlations with SalePrice:')
print(top_corr.head(20))

```



```
plt.figure(figsize=(10,8))
sns.heatmap(corr, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

■ **Output:**

Top 20 Correlations with SalePrice:

SalePrice 1.000000

OverallQual 0.790982

GrLivArea 0.708624

GarageCars 0.640409

GarageArea 0.623431

TotalBsmntSF 0.613581

1stFlrSF 0.605852

FullBath 0.560664

TotRmsAbvGrd 0.533723

YearBuilt 0.522897

The Pearson correlation matrix is computed using `df.corr(numeric_only=True)` which restricts computation to numeric columns. The correlations are then sliced to show only the `SalePrice` column, and sorted in descending order.

Key findings: OverallQual has the strongest linear relationship with SalePrice ($r = 0.791$), confirming that overall build quality is the dominant driver of value. GrLivArea ($r = 0.709$) is the second strongest — larger homes sell for more. GarageCars and GarageArea are highly correlated with each other (multicollinearity) and moderately correlated with price. The full correlation heatmap reveals additional structure, including strong inter-feature correlations among basement area metrics.

4.7 Feature-vs-Price Visualizations

Three targeted visualizations were created to illustrate the most important feature-price relationships:

Overall Quality vs SalePrice (Box Plot)

■ **Code:**

```
sns.boxplot(x='OverallQual', y='SalePrice', data=df)
plt.title('OverallQual vs SalePrice')
plt.show()
```

A box plot is ideal here because OverallQual is an ordinal integer (1–10). Each box shows the median, IQR, and outliers for homes of that quality rating. The pattern is unmistakable: as quality increases from 1 to 10, both the median price and the spread increase substantially. Quality 10 homes have medians above \$400,000.

Figure 4.2: Box plot of Overall Quality vs SalePrice. Higher quality ratings correspond strongly to higher prices.

Living Area vs SalePrice (Scatter Plot)

■ **Code:**

```
sns.scatterplot(x='GrLivArea', y='SalePrice', data=df)
```

```
plt.title('GrLivArea vs SalePrice')  
plt.show()
```

The scatter plot shows a strong positive linear trend: as above-grade living area increases, so does SalePrice. A few extreme outliers are visible — very large homes sold for surprisingly low prices, possibly due to distressed sales or unusual circumstances. These outliers are handled in preprocessing via IQR clipping.

Figure 4.3: Scatter plot of GrLivArea vs SalePrice showing strong positive linear relationship. Outliers in the upper-left corner are flagged for treatment.

Garage Capacity vs SalePrice (Box Plot)

■ Code:

```
sns.boxplot(x='GarageCars', y='SalePrice', data=df)  
plt.title('GarageCars vs SalePrice')  
plt.show()
```

Homes with more garage capacity (measured in car spaces) command higher prices. The step from 0 to 1 car garage produces the largest median price jump, while the difference between 3 and 4 spaces is modest. This diminishing return pattern suggests garage capacity is an important but non-linear feature.

5. Data Preprocessing

Data preprocessing transforms the raw dataset into a clean, numerical format suitable for linear regression. This section follows the notebook's preprocessing pipeline step by step, explaining every decision and its mathematical or practical justification.

5.1 Feature Selection Based on Correlation

■ Code:

```
corr = df.corr(numeric_only=True)
top_corr = corr['SalePrice'].sort_values(ascending=False).reset_index()
print('Top 15 Correlations with SalePrice:')
print(top_corr.head(16))
numeric_data = top_corr.head(16).iloc[:,0].tolist()
```

■ Output:

Top 15 Correlations with SalePrice:

index SalePrice

0 SalePrice 1.000000

1 OverallQual 0.790982

2 GrLivArea 0.708624

3 GarageCars 0.640409

4 GarageArea 0.623431

5 TotalBsmtSF 0.613581

6 1stFlrSF 0.605852

7 FullBath 0.560664

8 TotRmsAbvGrd 0.533723

9 YearBuilt 0.522897

Instead of using all 80 potential predictors, the top 15 numeric features by Pearson correlation with SalePrice are selected as the numeric feature set. Additionally, five categorical features are selected based on domain knowledge: Neighborhood, HouseStyle, Foundation, KitchenQual, and ExterQual. The total working dataset is reduced to 21 columns, which improves model interpretability and reduces noise from irrelevant features.

■ Code:

```
all_data = cat_features_all + numeric_data
df = df[all_data]
print(f'rows : {df.shape[0]} || columns : {df.shape[1]}')
```

■ Output:

data after dropping unnecessary columns:

rows : 1460 || columns : 21

5.2 Train/Test Split

■ Code:

```
X = df.drop('SalePrice', axis=1)
y = df['SalePrice']
x_tr, x_test, y_tr, y_test = train_test_split(
X, y, test_size=0.2, random_state=40)
print(f'shape of train data {x_tr.shape}')
print(f'shape of test data {x_test.shape}')
```

■ Output:

```
shape of train data (1168, 20)
shape of test data (292, 20)
```

The dataset is split into training (80%) and test (20%) subsets using a fixed `random_state=40` for reproducibility. This yields 1,168 training samples and 292 test samples. The split is performed *before* any imputation or scaling to prevent data leakage — statistics for imputation and scaling are computed only on training data and then applied to the test set.

5.3 Missing Value Imputation

Missing values are handled differently based on data type and distribution:

- **Median imputation** for continuous numeric columns (LotFrontage, MasVnrArea, GarageYrBlt, GrLivArea, GarageArea, TotalBsmtSF, 1stFlrSF, BsmtFinSF1, YearBuilt, YearRemodAdd). Median is preferred over mean because it is robust to the right-skewed distributions observed in the EDA.
- **Mode imputation** for discrete count-like columns (GarageCars, FullBath, TotRmsAbvGrd, Fireplaces, OverallQual). These columns represent whole-number counts where the most frequent value is the most logical substitute.

■ Code:

```
simple_median = SimpleImputer(strategy='median')
x_tr[median_cols] = simple_median.fit_transform(x_tr[median_cols])
x_test[median_cols] = simple_median.transform(x_test[median_cols])

simple_mode = SimpleImputer(strategy='most_frequent')
x_tr[mode_cols] = simple_mode.fit_transform(x_tr[mode_cols])
x_test[mode_cols] = simple_mode.transform(x_test[mode_cols])
```

The critical pattern here is `fit_transform` on training data and `transform` (without refitting) on test data. This ensures that the imputation values computed from training statistics are consistently applied to the test set, simulating real-world deployment.

5.4 Categorical Encoding

Two encoding strategies are applied based on whether the categorical variable has an inherent order:

Ordinal Encoding — KitchenQual and ExterQual

Both KitchenQual and ExterQual represent quality ratings with a clear ordering: Fa (Fair) < TA (Typical/Average) < Gd (Good) < Ex (Excellent). Ordinal encoding maps these to integers 0–3, preserving the rank relationship. This is crucial — using One-Hot Encoding here would destroy the ordering information.

■ **Code:**

```
kitchen_andExter_order = [['Fa', 'TA', 'Gd', 'Ex'], ['Fa', 'TA', 'Gd', 'Ex']]
ord_encoder = OrdinalEncoder(categories=kitchen_andExter_order)
x_tr[['KitchenQual', 'ExterQual']] = ord_encoder.fit_transform(
    x_tr[['KitchenQual', 'ExterQual']])
x_test[['KitchenQual', 'ExterQual']] = ord_encoder.transform(
    x_test[['KitchenQual', 'ExterQual']])
```

One-Hot Encoding — Neighborhood, HouseStyle, Foundation

For nominal categories without inherent order, One-Hot Encoding (OHE) is applied. Each unique category becomes a binary (0/1) indicator column. This avoids imposing a false numerical ordering on unordered categories.

■ **Code:**

```
x_tr = pd.get_dummies(x_tr, columns=cat_features_ohe, dtype=int)
x_test = pd.get_dummies(x_test, columns=cat_features_ohe, dtype=int)
x_test = x_test.reindex(columns=x_tr.columns, fill_value=0)
print('shape after OHE:')
print(x_tr.shape)
print(x_test.shape)
```

■ **Output:**

```
shape after OHE:
(1168, 56)
(292, 56)
```

After OHE, the feature space expands from 20 to **56 columns**. The `reindex` step is essential: if a category appears in the training set but not the test set (or vice versa), column alignment ensures no `KeyError` during prediction. Missing columns are filled with 0.

5.5 Outlier Treatment (IQR Method)

Outliers in numeric features are detected and removed using the Interquartile Range (IQR) method. For each numeric column, values beyond $1.5 \times \text{IQR}$ below Q_1 or above Q_3 are identified as outliers and *clipped* (Winsorized) rather than deleted — this preserves the row count while limiting the influence of extreme values.

$$\text{IQR} = Q_3 - Q_1 \mid \text{Lower} = Q_1 - 1.5 \times \text{IQR} \mid \text{Upper} = Q_3 + 1.5 \times \text{IQR}$$

■ **Code:**

```
for i in numeric_data:
    if i != 'SalePrice':
        Q1 = x_tr[i].quantile(0.25)
        Q3 = x_tr[i].quantile(0.75)
```

```
IQR = Q3 - Q1
up_bound = Q3 + 1.5 * IQR
low_bound = Q1 - 1.5 * IQR
x_tr[i] = x_tr[i].clip(low_bound, up_bound).astype(int)
x_test[i] = x_test[i].clip(low_bound, up_bound).astype(int)
```

■ **Output:**

```
outliers in OverallQual column: 2
outliers in GrLivArea column: 21
(clipped to boundary values, rows retained)
```

5.6 Feature Scaling (StandardScaler)

■ **Code:**

```
std_scaler = StandardScaler()
x_tr[scaled_cols] = std_scaler.fit_transform(x_tr[scaled_cols])
x_test[scaled_cols] = std_scaler.transform(x_test[scaled_cols])
```

StandardScaler transforms each feature by subtracting its training mean and dividing by its training standard deviation, producing a zero-mean, unit-variance distribution:

$$\mathbf{x}_{\text{scaled}} = (\mathbf{x} - \text{mean}) / \text{std}$$

Scaling is applied only to the median-imputed continuous columns (those with large value ranges such as LotFrontage, GrLivArea, TotalBsmtSF). The binary OHE columns and small-range mode-imputed columns (GarageCars: range 0–4, FullBath: 0–3) are left unscaled, as scaling them would provide no benefit and might harm interpretability.

Scaling is critical for Gradient Descent: without it, features with large magnitudes dominate the gradient computation and the algorithm may diverge or converge extremely slowly. The Normal Equation and Statistical approaches are theoretically scale-invariant, but consistent preprocessing ensures fair comparison.

6. Model Implementation

This project implements three regression engines, each representing a fundamentally different approach to parameter estimation. All implementations use only NumPy, without any sklearn regression classes.

6.1 Multiple Linear Regression — Gradient Descent (ML View)

6.1.1 Mathematical Foundation

The hypothesis function for multiple linear regression with p features is:

$$\hat{y} = B_0 + B_1x_1 + B_2x_2 + \dots + B_px_p$$

In vectorized form, using design matrix X ($n \times m$) with a bias column of ones prepended:

$$\hat{y} = X \cdot W + B_0$$

The cost function to minimise is the Mean Squared Error (MSE):

$$J(W, B_0) = (1/n) \sum (y_i - \hat{y}_i)^2$$

Gradient Descent updates the parameters iteratively in the direction of steepest descent of J . The update rules are:

$$W := W - \alpha \cdot (\partial J / \partial W) \text{ where } \partial J / \partial W = (2/n) \cdot X^T \cdot (\hat{y} - y)$$

$$B_0 := B_0 - \alpha \cdot (\partial J / \partial B_0) \text{ where } \partial J / \partial B_0 = (2/n) \cdot \sum (\hat{y} - y)$$

Here α (alpha) is the learning rate — a hyperparameter controlling step size. Too large a value causes divergence; too small causes very slow convergence. A tolerance threshold on the change in MSE allows early stopping when convergence is achieved.

6.1.2 Code Explanation — Class Structure

■ Code:

```
class MultipleLinearRegressionGD:
    def __init__(self):
        self.W_0 = None # Bias term (intercept)
        self.W = None # Weight vector (one per feature)
        self.SSE = float('inf') # Sum of Squared Errors
        self.MSE = None # Mean Squared Error
        self.loss_history = [] # Track loss over epochs
```

The constructor initialises all model parameters to None or infinity. `loss_history` stores the MSE at each epoch, enabling convergence plot generation.

■ Code:

```
def fit(self, X, y, learning_rate=0.01, epochs=1000,
        tolerance=1e-6, regularization=0, lambda_=0.1):
    self.X = np.array(X)
```

```

self.y = np.array(y)
self.n = len(self.y) # Number of samples
self.m = self.X.shape[1] # Number of features
self.B_0 = 0
self.B = np.zeros(self.m) # Initialise weights to zero
for epoch in range(epochs):
    y_hat = self.X @ self.B + self.B_0 # Matrix multiply
    error = y_hat - self.y
    SSE = np.sum(error**2)
    MSE = SSE / self.n
    self.loss_history.append(MSE)

    # Gradient update
    dB = (2/self.n) * self.X.T @ error
    dB_0 = (2/self.n) * np.sum(error)

    # Apply regularization to weights (NOT bias)
    if regularization == 1: # L1 (Lasso)
        dB += lambda_ * np.sign(self.B)
    elif regularization == 2: # L2 (Ridge)
        dB += 2 * lambda_ * self.B
    self.B -= learning_rate * dB
    self.B_0 -= learning_rate * dB_0

    if abs(self.SSE - SSE) < tolerance:
        break
    self.SSE = SSE
    self.MSE = MSE

```

Line-by-line breakdown:

- X , y are converted to NumPy arrays for vectorized math.
- n , m store the number of samples and features respectively.
- $B_0 = 0$, $B = \text{zeros}(m)$: parameters initialised at zero — a neutral starting point.
- $y_{\text{hat}} = X @ B + B_0$: vectorized prediction using matrix multiplication (faster than a loop).
- $\text{error} = y_{\text{hat}} - y$: residual vector of length n .
- $\text{SSE} = \text{sum}(\text{error}^2)$: sum of squared errors.
- $\text{dB} = (2/n) * X.T @ \text{error}$: gradient of MSE with respect to weights — a matrix-vector product.
- $\text{dB}_0 = (2/n) * \text{sum}(\text{error})$: gradient of MSE with respect to bias.
- Regularization terms are added to dB only (never to the bias term, which has no penalty in standard practice).
- $B -= \text{lr} * \text{dB}$: gradient descent update step.
- Early stopping: if the SSE change is smaller than `tolerance`, training halts.

6.1.3 Training and Evaluation

■ Code:

```

model = MultipleLinearRegressionGD()
model.fit(x_tr, y_tr, 0.001, 1000)

```


■ Code:

```
y_pred = model.predict(x_test)
print(f'Intercept (B0): {round(model.B_0, 3)}')
print(f'Mean squared error: {round(model.MSE, 3)}')
```

■ Output:

```
Intercept (B0): 895.772
Weights (B): [ 8.13e+03 6.94e+03 1.72e+04 9.72e+03 4.70e+03 ...]
Mean squared error is: 1,226,826,987
```

The model was trained with a learning rate of 0.001 over 1,000 epochs. The learning rate was kept conservative to ensure stable convergence across all 56 features. The intercept of approximately \$895 reflects the baseline prediction before any feature information is considered, and the weight vector assigns a positive coefficient to each of the 56 features that correlate positively with price.

6.2 Simple Linear Regression — Numerical View (Normal Equation)

6.2.1 Mathematical Foundation

The Normal Equation provides an *analytical* (closed-form) solution to the linear regression problem — no iterative steps required. For simple linear regression (one predictor x , target y), the sum of squared errors is minimised by solving the system of normal equations derived by setting the partial derivatives to zero:

$$\begin{aligned}\partial \text{SSE} / \partial a_0 &= 0 \rightarrow n a_0 + a_1 \sum x = \sum y \\ \partial \text{SSE} / \partial a_1 &= 0 \rightarrow a_0 \sum x + a_1 \sum x^2 = \sum xy\end{aligned}$$

In matrix form, this system is:

$$\begin{bmatrix} n & \sum x \\ \sum x & \sum x^2 \end{bmatrix} \cdot \begin{bmatrix} a_0 & a_1 \end{bmatrix}^T = \begin{bmatrix} \sum y & \sum xy \end{bmatrix}^T$$

Solving this 2×2 linear system directly using NumPy's `np.linalg.solve()` gives the optimal intercept a_0 and slope a_1 without any iteration.

6.2.2 Code Explanation

■ Code:

```
class SimpleLinearRegressionNumericalView:
    def fit(self, X, y, regularization=0, lambda_=0.1, show_steps=True):
        X = np.asarray(X, dtype=float).reshape(-1)
        y = np.asarray(y, dtype=float).reshape(-1)
        n = len(X)

        # Compute required summations
        sum_x = np.sum(X)
        sum_y = np.sum(y)
        sum_x2 = np.sum(X**2)
        sum_xy = np.sum(X * y)

        # Build the 2x2 normal equation matrix
        A = np.array([[n, sum_x],
                      [sum_x, sum_x2]])
```

```

b = np.array([sum_y, sum_xy])
# Apply regularization if requested
if regularization == 2: # Ridge: add lambda to diagonal
    A[1,1] += lambda_
# Solve the normal equations
solution = np.linalg.solve(A, b)
self.a0 = solution[0] # Intercept
self.a1 = solution[1] # Slope

```

The four summations ($\sum x$, $\sum y$, $\sum x^2$, $\sum xy$) are computed in a single vectorized pass over the data. They are assembled into the 2×2 matrix A and right-hand-side vector b . `np.linalg.solve(A, b)` uses Gaussian elimination to solve the system exactly. L2 regularization is implemented by adding λ to the bottom-right element of A (corresponding to the coefficient of the slope term), which corresponds to the Ridge normal equation: $(X^T X + \lambda I)^{-1} X^T y$.

■ Code:

```

feature_name = 'GrLivArea'
NumericalModel = SimpleLinearRegressionNumericalView()
NumericalModel.fit(x_tr[feature_name], y_tr)

```

■ Output:

Step 1: Define SSE (Sum of Squared Errors)

$SSE = \sum (y_i - a_0 - a_1 x_i)^2$

Step 2: Compute Summations

$n = 1168$

$\sum x_i = 0.00$

$\sum y_i = 211,864,417.00$

$\sum x_i^2 = 1,168.00$

$\sum x_i y_i = 66,879,526.81$

Step 3: Solve 2×2 Normal Equation System

→ a_0 (Intercept): 181,390.77

→ a_1 (Slope): 57,259.87

The model prints an instructive step-by-step walkthrough of the derivation. Since `GrLivArea` was already standardized (mean ≈ 0 , std = 1), $\sum x_i \approx 0$ and $\sum x_i^2 \approx n = 1168$. This makes the normal equations particularly clean — $a_0 \approx \bar{y}$ and $a_1 = \sum xy / n$.

6.3 Simple Linear Regression — Statistical View

6.3.1 Mathematical Foundation

The statistical view derives the optimal slope and intercept directly from sample statistics — no matrix algebra required. Given data $\{(x_i, y_i)\}$, the least-squares estimates are:

$$B_1 = SXY / SXX$$

$$B_0 = \bar{y} - B_1 \cdot \bar{x}$$

Where:

- $\bar{x}$ and $\bar{y}$ are the sample means of x and y respectively.
- $SXX = \sum (x_i - \bar{x})^2$ — the sum of squared deviations of x (numerically equivalent to $n \cdot \text{Var}(x)$).
- $SXY = \sum (x_i - \bar{x})(y_i - \bar{y})$ — the sample covariance (unnormalized).

The slope B_1 is therefore the ratio of the co-variation between x and y to the total variation of x — precisely the definition of the standardized regression coefficient in the simple case.

6.3.2 Code Explanation

■ Code:

```
class SimpleLinearRegressionStatisticalView:
    def fit(self, X, y, regularization=0, lambda_=0.1):
        X = np.array(X)
        y = np.array(y)

        x_bar = X.mean() # Sample mean of X
        y_bar = y.mean() # Sample mean of y

        SXX = np.sum((X - x_bar)**2) # Sum of squared deviations
        SXY = np.sum((X - x_bar) * (y - y_bar)) # Sum of cross-deviations

        match regularization:
            case 0: # No regularization
                self.B_1 = SXY / SXX
                self.B_0 = y_bar - self.B_1 * x_bar
            case 2: # Ridge: shrink slope denominator
                self.B_1 = SXY / (SXX + lambda_)
                self.B_0 = y_bar - self.B_1 * x_bar
            case 1: # Lasso: soft-thresholding on slope
                raw_B1 = SXY / SXX
                self.B_1 = np.sign(raw_B1) * max(abs(raw_B1) - lambda_, 0)
                self.B_0 = y_bar - self.B_1 * x_bar
```

The match statement (Python 3.10+ structural pattern matching) cleanly handles the three regularization cases. For Ridge (L2), the denominator SXX is augmented by λ , shrinking the slope toward zero. For Lasso (L1), soft-thresholding is applied: the slope is pushed toward zero by λ , and set to exactly zero if $|B_1| \leq \lambda$.

■ Code:

```
feature = 'GrLivArea'
stat_model = SimpleLinearRegressionStatisticalView()
stat_model.fit(x_tr[feature], y_tr)
print(f'Line of best fit: y = {stat_model.B_0:.3f} + {stat_model.B_1:.3f}x')
print(f'MSE: {stat_model.MSE:.3f}')
```

■ Output:

```
Training Simple Linear Regression for: GrLivArea vs SalePrice
Line of best fit: y = 181390.768 + 57259.869x
MSE: 3,318,712,587.870
```

The results match the Numerical View almost exactly (181,390.768 vs 181,390.77, 57,259.869 vs 57,259.87) — as expected, since both methods solve the same least-squares problem. The tiny difference is due to floating-point arithmetic in different computation paths.

6.4 Regularization (L1 Lasso and L2 Ridge)

Regularization modifies the cost function by adding a penalty term that discourages large coefficient values, addressing overfitting in high-dimensional settings.

6.4.1 L2 Ridge Regularization

$$J_{\text{Ridge}}(\mathbf{w}) = (1/n) \sum (y - \hat{y})^2 + \lambda \cdot \sum \mathbf{B}_j^2$$

Ridge adds the sum of squared weights multiplied by λ to the cost function. The effect is that the gradient update includes an additional shrinkage term: $\text{dB} += 2 * \text{lambda_} * \text{self.B}$. This continuously pushes all weights toward zero proportionally to their magnitude, but never sets them exactly to zero. Ridge is particularly effective when many features contribute small but non-zero effects.

6.4.2 L1 Lasso Regularization

$$J_{\text{Lasso}}(\mathbf{w}) = (1/n) \sum (y - \hat{y})^2 + \lambda \cdot \sum |\mathbf{B}_j|$$

Lasso adds the sum of absolute weight values. The gradient term is $\text{dB} += \text{lambda_} * \text{np.sign}(\text{self.B})$, where np.sign returns +1 or -1 depending on the sign of each weight. This constant-magnitude penalty can drive small coefficients to exactly zero, effectively performing feature selection — a crucial advantage for high-dimensional problems.

6.4.3 Comparison

Property	Ridge (L2)	Lasso (L1)
Penalty term	$\lambda \cdot \sum \mathbf{B}_j^2$	$\lambda \cdot \sum \mathbf{B}_j $
Effect on weights	Shrinks toward zero, never exactly zero	Can set weights exactly to zero
Feature selection	No	Yes (implicit)
Best when	Many small predictive features	Sparse true model (few relevant features)
Gradient update	$+2\lambda \mathbf{B}_j$	$+\lambda \cdot \text{sign}(\mathbf{B}_j)$

Table 6.1: Comparison of L1 (Lasso) and L2 (Ridge) regularization

7. Model Evaluation and Comparison

Three model specifications were evaluated, corresponding to three feature selection strategies. All three use sklearn's LinearRegression for fair sklearn-based comparison alongside the custom implementations.

7.1 Evaluation Metrics

Mean Squared Error (MSE)

$$\text{MSE} = (1/n) \sum (y_i - \hat{y}_i)^2$$

MSE measures the average squared prediction error. It penalises large errors disproportionately (due to squaring), making it sensitive to outliers. Lower is better. Its units are the square of the target variable (USD²).

Root Mean Squared Error (RMSE)

$$\text{RMSE} = \sqrt{\text{MSE}}$$

RMSE is the square root of MSE, returning the error to the original units of the target (USD). It is easier to interpret — e.g., an RMSE of \$35,000 means the model is off by about \$35,000 on average.

Coefficient of Determination (R²)

$$R^2 = 1 - \text{SSE} / \text{SST} \text{ where } \text{SST} = \sum (y_i - \bar{y})^2$$

R² measures the proportion of variance in the target explained by the model. A value of 0 means the model is no better than predicting the mean; a value of 1 means perfect prediction. Values between 0.7 and 0.9 are considered strong for housing data.

7.2 Three Model Specifications

Model 1 — Best Single Predictor (GrLivArea)

■ Code:

```
model1_features = ['GrLivArea']
lr_model1 = LinearRegression()
lr_model1.fit(x_tr[model1_features], y_tr)
```

■ Output:

R² (Train): 0.4974

R² (Test): 0.5294

Intercept: \$181,390.77

Coefficient (GrLivArea): 57,259.87

A single-feature model using only GrLivArea explains about 50% of the variance in house prices. Each additional unit of standardized living area adds ~\$57,260 to the predicted price. The test R²

(0.5294) slightly exceeds training R^2 (0.4974), which can occur by chance with small test sets and suggests no overfitting — a single predictor is inherently parsimonious.

Model 2 — Full Model (56 Numeric Features)

■ Code:

```
numeric_features = x_tr.select_dtypes(include=[np.number]).columns.tolist()
lr_model2 = LinearRegression()
lr_model2.fit(x_tr[numeric_features], y_tr)
```

■ Output:

R^2 (Train): 0.8503

R^2 (Test): 0.7947

Intercept: \$59,392.83

Top coefficient: Neighborhood_NoRidge: 65,008.78

Using all 56 engineered features dramatically improves explanatory power. The model explains 85% of training variance and 79.5% of test variance. The gap between train and test R^2 (0.0556) indicates mild overfitting — the model has learned some training-specific patterns. The top coefficients are one-hot encoded neighborhood dummies (NoRidge, StoneBr, NridgHt), confirming that location is a major price driver.

Model 3 — Domain Knowledge Selection (10 Features)

■ Code:

```
domain_features = ['GrLivArea', 'GarageArea', 'TotalBsmtSF', '1stFlrSF',
                  'FullBath', 'YearBuilt', 'YearRemodAdd', 'MasVnrArea',
                  'BsmtFinSF1', 'HouseStyle_SFoyer']
lr_model3 = LinearRegression()
lr_model3.fit(x_tr[domain_features], y_tr)
```

■ Output:

R^2 (Train): 0.7593

R^2 (Test): 0.7750

Intercept: \$183,052.52

Top coefficient: GrLivArea: 36,543.49

The domain knowledge model selects 10 features based on housing expertise: size metrics (GrLivArea, GarageArea, TotalBsmtSF, 1stFlrSF, BsmtFinSF1, MasVnrArea), amenity quality (FullBath), temporal factors (YearBuilt, YearRemodAdd), and architectural style (HouseStyle_SFoyer). This model achieves test R^2 of 0.7750 — only 2 percentage points below the full model, while using just 10/56 features. Notably, the train-test gap is only 0.0157, indicating much better generalisation than the full model.

7.3 Summary Comparison Table

Model	Features	R^2 Train	R^2 Test	Train-Test Gap
Model 1: Single Predictor	1 (GrLivArea)	0.4974	0.5294	−0.032

Model 2: Full Model	56 features	0.8503	0.7947	+0.056
Model 3: Domain Knowledge	10 features	0.7593	0.7750	−0.016

Table 7.1: Model comparison summary. Negative gap indicates slight generalisation advantage on test set; positive indicates mild overfitting. Model 3 offers the best efficiency (high R^2 with few features).

7.4 Residual Analysis

Residual plots were generated for all three models by plotting predicted values on the x-axis against residuals (actual – predicted) on the y-axis. Key observations:

- **Model 1 residuals** show a funnel shape (heteroscedasticity): residual variance increases as predicted price increases, suggesting the single feature cannot capture non-linear patterns in higher-value homes.
- **Model 2 residuals** are more uniformly distributed around zero across the prediction range, indicating better variance capture, though a few outliers remain.
- **Model 3 residuals** are intermediate — more homoscedastic than Model 1 but with slightly wider spread than Model 2.

Residual standard deviations: Model 1 $\approx$ \$57,584; Model 2 $\approx$ \$31,426; Model 3 $\approx$ \$39,848. These confirm that the full model achieves the tightest predictions.

8. Interactive Visualizations (Plotly)

Section 6 of the notebook implements eight interactive visualizations using Plotly — a library that generates interactive HTML/JavaScript charts with hover tooltips, zoom controls, and filtering capabilities. Additionally, Task 2.5 implements five specific required visualizations. All charts were generated on the raw dataset (`df_viz`) for maximum completeness.

■ Code:

```
import plotly.graph_objects as go
import plotly.express as px
from plotly.subplots import make_subplots
df_viz = pd.read_csv('../data/House Prices Dataset.csv')
```

■ Output:

```
Data loaded: 1460 rows × 81 columns
```

Visualization 1: Dynamic Correlation Explorer

An interactive horizontal bar chart showing the Pearson correlation of all numeric features with `SalePrice`. Bars are color-coded using a Red-Yellow-Green gradient (red for negative, green for positive correlation). The chart confirms that `OverallQual` (0.791) and `GrLivArea` (0.709) are the strongest positive predictors, while features like `MiscVal` and `YrSold` show near-zero correlations. The interactive tooltip reveals exact correlation values on hover.

■ Code:

```
corr_with_price = df_viz.corr(numeric_only=True)['SalePrice'].drop('SalePrice')
corr_with_price = corr_with_price.sort_values()
fig1 = go.Figure()
fig1.add_trace(go.Bar(y=corr_with_price.index,
x=corr_with_price.values, orientation='h',
marker=dict(colorscale='RdYlGn', ...)))
fig1.show()
```

Visualization 2: Neighborhood Price Comparator

A violin plot comparing `SalePrice` distributions across all 25 neighborhoods, sorted by median price in descending order. The violin shape reveals not just the median but the full price distribution within each neighborhood. `NridgHt`, `NoRidge`, and `StoneBr` consistently produce the highest-priced homes (medians \$315K, \$301K, \$278K). `MeadowV` and `IDOTRR` are at the bottom with medians below \$100K.

■ Code:

```
for neighborhood in neighborhood_stats['Neighborhood']:
fig2.add_trace(go.Violin(y=neighborhood_data, name=neighborhood, ...))
fig2.show()
```

Visualization 3: Temporal Price Trend (2006–2010)

A dual-axis time series plot showing monthly mean price (bar), median price (line), and transaction count (area). The data spans January 2006 to July 2010. Key observations: housing prices were relatively stable during the period, with average monthly prices ranging from \$121,750 to \$223,769. Transaction volume peaked in spring/summer months (May–July) each year, consistent with seasonal housing market patterns.

■ **Code:**

```
temporal_agg = df_temporal.groupby('YearMonth').agg({'SalePrice':
['mean', 'median', 'count']})
fig3 = make_subplots(rows=2, cols=1, ...)
# Add mean bar and median line on row 1
# Add transaction count on row 2
fig3.show()
```

Visualization 4: Outlier Investigation Dashboard

An interactive scatter plot (GrLivArea vs SalePrice, colored by OverallQual) where statistical outliers are highlighted in red. The IQR method identifies 61 outliers (4.2% of homes), with prices outside the range \$3,938–\$340,038. Outlier properties tend to have high GrLivArea (mean 2,457 sq ft) and high quality (mean 8.75/10) but were sold at unusually high prices. The interactive lasso/box selection tool allows drilling into specific outlier clusters.

■ **Code:**

```
Q1, Q3 = df_outliers['SalePrice'].quantile([0.25, 0.75])
IQR = Q3 - Q1
df_outliers['IsOutlier'] = ((SalePrice < Q1-1.5*IQR) | (SalePrice > Q3+1.5*IQR))
fig4.show()
```

Visualization 5: Hierarchical Sunburst (Zone → Neighborhood → Style)

A drill-down sunburst chart that encodes the three-level hierarchy: MSZoning (e.g., RL-Residential Low, RM-Residential Medium) → Neighborhood → HouseStyle. Segment size represents transaction count; color represents median price. The chart enables rapid discovery of which combinations yield the highest-value properties.

■ **Code:**

```
fig5 = go.Figure(go.Sunburst(
labels=[...], parents=[...],
values=hierarchy_agg['Count'],
branchvalues='total', ...))
fig5.show()
```

Visualization 6: Feature Impact Playground

A scatter plot with marginal histograms (GrLivArea vs SalePrice), with bubble size proportional to GarageArea and color proportional to SalePrice. The correlation matrix confirms GrLivArea–SalePrice (0.709), OverallQual–SalePrice (0.791), and strong inter-feature correlations among size metrics.

■ **Code:**

```
fig6 = px.scatter(df_impact, x='GrLivArea', y='SalePrice',
```

```
color='SalePrice', size='GarageArea',
marginal_x='histogram', marginal_y='histogram')
fig6.show()
```

Visualization 7: Parallel Coordinates Chart

A parallel coordinates plot connects each house as a line across six axes: OverallQual, GrLivArea, GarageArea, YearBuilt, TotalBsmtSF, and SalePrice. Lines are colored by SalePrice (red = cheap, green = expensive). Users can brush each axis to isolate subsets — for example, filtering to houses with both high GrLivArea and high OverallQual reveals that these properties concentrate in the high-price green range.

■ Code:

```
fig7 = go.Figure(data=go.Parcoords(
    line=dict(color=df_parallel['SalePrice'], colorscale='RdYlGn'),
    dimensions=[dict(values=df_parallel[f]) for f in parallel_features]))
fig7.show()
```

Visualization 8: Missing Data Heatmap

An interactive heatmap where rows represent features with missing values and columns represent individual houses (first 50 rows sampled). Yellow cells indicate missing values; purple cells indicate present values. The visualization confirms that Pool-, Misc-, and Alley-related features are almost entirely absent, while Garage- and Basement-related fields have structured missingness (present only in houses that have those amenities).

■ Code:

```
missing_data = df_viz.isnull().astype(int)
df_missing_sample = missing_data[missing_cols.index].head(50)
fig8 = go.Figure(data=go.Heatmap(z=df_missing_sample.values.T, ...))
fig8.show()
```

Task 2.5: Required Project Visualizations

1. Full Correlation Heatmap (38 Numeric Variables)

■ Code:

```
numeric_cols = df_viz.select_dtypes(include=[np.number]).columns.tolist()
corr_matrix = df_viz[numeric_cols].corr()
fig_corr_heatmap = go.Figure(data=go.Heatmap(
    z=corr_matrix.values, x=numeric_cols, y=numeric_cols,
    colorscale='RdBu', zmid=0, zmin=-1, zmax=1))
fig_corr_heatmap.show()
```

■ Output:

```
Correlation matrix shape: (38, 38)
Numeric features analyzed: 38
```

A 38×38 correlation heatmap covering all numeric variables. Blue cells indicate positive correlation; red cells indicate negative. The heatmap reveals several strong multi-collinearity clusters: (1stFlrSF,

TotalBsmtSF), (GarageArea, GarageCars), (YearBuilt, YearRemodAdd), and (TotRmsAbvGrd, GrLivArea).

2. Regression Line Scatter Plot

■ Code:

```
# Fit sklearn regression for plotting
lr_viz = LinearRegression().fit(X_clean, y_clean)
y_pred_viz = lr_viz.predict(X_sorted.reshape(-1,1))
fig_scatter = go.Figure()
fig_scatter.add_trace(go.Scatter(x=GrLivArea, y=SalePrice, mode='markers'))
fig_scatter.add_trace(go.Scatter(x=X_sorted, y=y_pred_viz, mode='lines'))
fig_scatter.show()
```

■ Output:

```
Regression equation: SalePrice = 18,569.03 + 107.13 × GrLivArea
R² score: 0.5021
```

Note: This plot uses unstandardized GrLivArea values from the raw dataset (not the scaled training data), which explains the different slope (107.13) compared to the scaled version (57,259.87). The R^2 of 0.5021 is consistent with both implementations.

3. Gradient Descent Convergence Plot

■ Code:

```
gd_tracker = GDTracker()
gd_tracker.fit(x_tr, y_tr, learning_rate=0.001, epochs=1000)
fig_conv = go.Figure()
fig_conv.add_trace(go.Scatter(x=list(range(1000)), y=gd_tracker.loss_history))
fig_conv.show()
```

■ Output:

```
Total iterations: 1000
Initial SSE: 31,410,840,327,402
Final SSE: 1,432,933,921,234
Reduction: 95.44%
Final MSE: 1,226,826,987
```

The convergence curve shows rapid initial improvement (MSE drops steeply in the first ~100 epochs) followed by gradual flattening. The 95.44% SSE reduction demonstrates that gradient descent with $lr=0.001$ is effective for this problem. The curve is monotonically decreasing — no oscillations — confirming the learning rate is in the stable range.

4 & 5. Residual Plots and Coefficient Bar Charts

Residual plots were discussed in detail in Section 7.4. The coefficient bar chart compares the top feature coefficients across all three model specifications, making visible how the full model (Model 2) assigns large coefficients to premium neighborhood dummies, while the domain knowledge model (Model 3) concentrates most weight on physical size features (GrLivArea, TotalBsmtSF) and temporal factors (YearBuilt).

9. Conclusion

9.1 Summary of Findings

This project successfully implemented and compared three algorithmic approaches to linear regression on the Ames Housing dataset. The key findings are:

- All three engines (Gradient Descent, Normal Equation, Statistical formulas) converged to identical or near-identical parameter estimates for the same simple linear regression problem, validating the correctness of the from-scratch implementations.
- The full model (Model 2, 56 features) achieved the highest test R^2 of 0.7947, explaining nearly 80% of variance in house prices. The domain knowledge model (Model 3, 10 features) was only marginally weaker ($R^2 = 0.7750$) while being far more interpretable and generalizable.
- Overall quality (OverallQual) and above-grade living area (GrLivArea) are the two single strongest predictors, with Pearson correlations of 0.791 and 0.709 respectively. Premium neighborhood membership (NoRidge, StoneBr, NridgHt) adds \$39,000–\$65,000 to predicted prices in the full model.
- L1 and L2 regularization were integrated into all custom models. L1 (Lasso) drives small coefficients to zero, providing implicit feature selection. L2 (Ridge) smoothly shrinks all coefficients, reducing variance at the cost of slight bias.
- The gradient descent convergence curve showed 95.44% SSE reduction over 1,000 epochs with $lr=0.001$, confirming stable and effective optimisation.

9.2 Limitations

- **Linearity assumption:** Linear regression assumes the relationship between features and price is linear. The right-skewed SalePrice distribution and heteroscedastic residuals suggest a log-transformation of the target could improve model fit.
- **Feature engineering:** Only basic categorical encoding was applied. Interaction terms (e.g., OverallQual $\times$ GrLivArea) and polynomial features could capture non-linear patterns.
- **Gradient Descent hyperparameters:** The learning rate (0.001) and epoch count (1,000) were not tuned systematically. A learning rate scheduler or Adam optimiser could improve convergence.
- **Spatial effects:** The neighborhood one-hot encoding is a blunt instrument. Geographic regression or spatial autocorrelation models could capture location effects more precisely.

9.3 Future Work

- Implement polynomial regression and interaction features to capture non-linear relationships.
- Apply Elastic Net regularization (combination of L1 and L2) for optimal coefficient shrinkage.
- Implement k-fold cross-validation for more robust model evaluation.
- Experiment with log-transformation of SalePrice to reduce skewness and improve residual normality.
- Extend to ensemble methods (Random Forest, Gradient Boosting) as a comparison baseline.

10. Appendix — Full Notebook Code Reference

This appendix contains all code implemented across the notebook cells, presented in the original order for complete reference.

A.1 Complete Gradient Descent Implementation

■ Code:

```
class MultipleLinearRegressionGD:
    def __init__(self):
        self.W_0 = None
        self.W = None
        self.SSE = float('inf')
        self.MSE = None
        self.loss_history = []

    def fit(self, X, y, learning_rate=0.01, epochs=1000,
            tolerance=1e-6, regularization=0, lambda_=0.1):
        self.X = np.array(X)
        self.y = np.array(y)
        self.n = len(self.y)
        self.m = self.X.shape[1]
        self.B_0 = 0
        self.B = np.zeros(self.m)
        for epoch in range(epochs):
            y_hat = self.X @ self.B + self.B_0
            error = y_hat - self.y
            SSE = np.sum(error**2)
            MSE = SSE / self.n
            self.loss_history.append(MSE)
            dB = (2/self.n) * self.X.T @ error
            dB_0 = (2/self.n) * np.sum(error)
            if regularization == 1:
                dB += lambda_ * np.sign(self.B)
            elif regularization == 2:
                dB += 2 * lambda_ * self.B
            self.B -= learning_rate * dB
            self.B_0 -= learning_rate * dB_0
            if abs(self.SSE - SSE) < tolerance:
                break
            self.SSE = SSE
            self.MSE = MSE

    def predict(self, X):
        X = np.array(X)
```

```

return X @ self.B + self.B_0

def plot(self, X_test, y_test):
    y_pred = self.predict(X_test)
    plt.figure(figsize=(8,5))
    plt.scatter(y_test, y_pred, alpha=0.5)
    plt.xlabel('Actual'); plt.ylabel('Predicted')
    plt.title('Actual vs Predicted (GD Model)')
    plt.plot([y_test.min(), y_test.max()],
             [y_test.min(), y_test.max()], 'r--')
    plt.show()

```

A.2 Complete Numerical (Normal Equation) Implementation

■ Code:

```

class SimpleLinearRegressionNumericalView:
    def __init__(self):
        self.a0 = None
        self.a1 = None
        self.SSE = None
        self.MSE = None

    def predict(self, X):
        X = np.asarray(X, dtype=float).reshape(-1)
        return self.a0 + self.a1 * X

    @staticmethod
    def sse(y, y_hat):
        return np.sum((y - y_hat) ** 2)

    def fit(self, X, y, regularization=0, lambda_=0.1, show_steps=True):
        X = np.asarray(X, dtype=float).reshape(-1)
        y = np.asarray(y, dtype=float).reshape(-1)
        n = len(X)
        sum_x = np.sum(X)
        sum_y = np.sum(y)
        sum_x2 = np.sum(X**2)
        sum_xy = np.sum(X * y)
        A = np.array([[n, sum_x], [sum_x, sum_x2]])
        b = np.array([sum_y, sum_xy])
        if regularization == 2:
            A[1,1] += lambda_
        solution = np.linalg.solve(A, b)
        self.a0 = solution[0]
        self.a1 = solution[1]
        y_hat = self.predict(X)
        self.SSE = self.sse(y, y_hat)
        self.MSE = self.SSE / n

```

A.3 Complete Statistical View Implementation

■ Code:

```
class SimpleLinearRegressionStatisticalView:
    def __init__(self):
        self.B_0 = None
        self.B_1 = None
        self.MSE = None

    def fit(self, X, y, regularization=0, lambda_=0.1):
        X = np.array(X)
        y = np.array(y)
        x_bar = X.mean()
        y_bar = y.mean()
        SXX = np.sum((X - x_bar)**2)
        SXY = np.sum((X - x_bar) * (y - y_bar))
        match regularization:
            case 0:
                self.B_1 = SXY / SXX
                self.B_0 = y_bar - self.B_1 * x_bar
            case 2:
                self.B_1 = SXY / (SXX + lambda_)
                self.B_0 = y_bar - self.B_1 * x_bar
            case 1:
                raw_B1 = SXY / SXX
                self.B_1 = np.sign(raw_B1) * max(abs(raw_B1) - lambda_, 0)
                self.B_0 = y_bar - self.B_1 * x_bar
        y_hat = self.predict(X)
        n = len(y)
        self.SSE = np.sum((y - y_hat)**2)
        self.MSE = self.SSE / n

    def predict(self, X):
        X = np.array(X)
        return self.B_0 + self.B_1 * X

    def plot(self, X, y):
        y_hat = self.predict(X)
        plt.figure(figsize=(8,5))
        plt.scatter(X, y, alpha=0.4, label='Actual')
        plt.plot(sorted(X), self.predict(sorted(X)), 'r-', label='Fitted')
        plt.xlabel('GrLivArea (scaled)')
        plt.ylabel('SalePrice ($)')
        plt.title('Simple Linear Regression: Statistical View')
        plt.legend()
        plt.show()
```

A.4 Model Specification Configurations

■ Code:

```
# Model 1: Best single predictor
model1_features = ['GrLivArea']

# Model 2: All numeric features (post-preprocessing)
numeric_features = x_tr.select_dtypes(include=[np.number]).columns.tolist()

# Model 3: Domain knowledge selection
domain_features = [
    'GrLivArea', 'GarageArea', 'TotalBsmtSF', '1stFlrSF',
    'FullBath', 'YearBuilt', 'YearRemodAdd', 'MasVnrArea',
    'BsmtFinSF1', 'HouseStyle_SFoyer'
]

# Train all three sklearn models for comparison
from sklearn.linear_model import LinearRegression
lr_model1 = LinearRegression().fit(x_tr[model1_features], y_tr)
lr_model2 = LinearRegression().fit(x_tr[numeric_features], y_tr)
lr_model3 = LinearRegression().fit(x_tr[domain_features], y_tr)
```

End of Report — Linear Regression From Scratch • Alexandria University • Spring 2026